

This is a logbook entry to complete and email to your faculty supervisor as progress update on a biweekly basis (i.e. once every 2 weeks). Your submissions will be used as reference for your internship progress and will be assessed towards the overall performance of your internship journal.

Important Note:

1. As consistency of submissions is assessed as part of your internship, all submissions should be made on time. Any delays should be informed and prior approval to be sought with your faculty supervisor. Late entries without prior approval will be considered invalid for assessment.
2. Before submission, please obtain clearance for your logbook content from your organisation supervisor to ensure that no confidential information is included.

Student Name	:	Tan Yi Rong
Student ID	:	U2320967K
Company Name	:	GlobalFoundries
Company Supervisor	:	Ng, Christopher
Logbook Entry Period	:	25/05/2025 to 07/05/2025
Date of Submission	:	09/09/2025

A. State the Internship Objective, Scope, and Problem Statement

(Briefly describe the objective(s) provided by the internship organisation, and the expected scope of work and deliverables expected from you to address the issues.)

Objective:

The objective of this internship is to support the Automated Material Handling System (AMHS) department in improving Overhead Hoist Transport (OHT) maintenance data collection and predictive maintenance requirements for parts replacement. This is intended to reduce error occurrences and improve operational performance during production.

Scope of Work:

- Understudy AMHS operation and OHT Preventive Maintenance (PM) process during the initial stage of the internship.
- Continuation of Project Odin, which focuses on OHT wheel measurement through sensor testing and data collection for Predictive Maintenance.
- Carry out preliminary industry-attached FYP scoping and research related to AMHS and OHT maintenance improvement.
- Attend team meetings, daily progress reviews and support documentation and other ad-hoc AMHS-related tasks where required.

Problem Statement:

The current OHT maintenance and condition monitoring process requires more effective and reliable methods of data collection, analysis and maintenance planning. Without sufficient monitoring and process improvement, maintenance activities may remain reactive, labour-intensive and more prone to inconsistency. This internship focuses on understanding current

workflow, supporting ongoing wheel monitoring and Predictive Maintenance (PdM) efforts and exploring ideas that may improve maintenance efficiency, safety and ease of line support work.

B. Describe your work done over the past 2 weeks

Work and/or Tasks Performed	Date (From)	Date (To)
Understudy AMHS operation and familiarize with the OHT Preventive Maintenance (PM) process	25/05/26	27/05/26
Observe line support work scope and identify current maintenance practices, challenges and safety considerations	27/05/26	03/06/26
Complete Sumtotal Security / access and AI quizzes.	25/05/26	29/05/26
Explore possible project and improvement ideas that may be relevant for FYP scoping and future implementations	03/06/26	03/06/26

C. Describe and reflect on your work/task outcome/learning over the last 2 weeks

(Illustrate the challenges that you are facing; describe areas that you have done well and areas that could be better managed)

This bi-weekly period was mainly focused on building foundational understanding of the AMHS working environment and the preventive maintenance process for the OHT system. In the first 2 weeks, my main priority was to familiarize myself with the line support workflow, observe how PM is carried out and understand the practical challenges faced during day-to-day PM work. This helped me build a stronger foundation before moving into the more technical parts of the internship, such as Project Odin and FYP-related research.

During this period, I also started thinking about possible improvement ideas that could make line support work easier or safer, while considering whether any of these ideas may be suitable for further development as an industry-attached FYP topic. Overall, these 2 weeks were useful for understanding the working environment and identifying early opportunities for improvement, although I am still in the process of building deeper technical knowledge.